

Star Formation Rates in Local Environments: Galaxy NGC 1566

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ABSTRACT

Understanding how stars form in molecular gas helps us view the broader picture of when, why, and how galaxies build up their stellar mass. Using H α to trace the star formation rate and CO(2-1) as a tracer for molecular gas to study the Kennicutt-Schmidt (KS) relationship at ~ 100 pc resolution in NGC 1566. We studied the KS relation at various spatial resolutions across environments of NGC 1566 and found that the KS slope changes systematically with resolution. We performed a pixel-to-pixel level analysis using Gaussian convolution to produce identical resolution in both H α and CO(2-1) distributions. Our results show that the correlation between H α and CO(2-1) intensities in NGC 1566 reports a global star formation law index (Σ_{SFR} vs. Σ_{gas}) of 0.92 ± 0.02 . Our analysis shows a shallower slope than the Kennicutt-Schmidt (Kennicutt 1998) literature value of 1.4 as we decomposed the galaxy into various local environments. We observed a slope of 1.04 ± 0.06 in the interarms, 0.36 ± 0.05 in the bar, and 0.78 ± 0.03 in the spiral arms. From these results, we can infer that the Kennicutt-Schmidt relation between Σ_{SFR} and Σ_{gas} is not universal at high pixel-to-pixel resolution for individual galaxy environments however, a more in-depth study with a larger sample of galaxies can gauge the effectiveness of this method.

Keywords: H alpha lines — Interstellar medium — Markov chain Monte Carlo — Molecular clouds — Star formation

1. INTRODUCTION

Star formation occurs in the cold, dense molecular gas regions of galaxies, which is a pivotal step in their evolution. A process that is governed by both large and small physics, including feedback from other newly formed stars, supernovae, and magnetic fields (Jeřábková et al. 2018), (Ostriker & Shetty 2011). As a result of these complex interactions and mechanisms, the relation between star formation and its contributing factors scale differently at varying galactic scales raising a fundamental question: how does the relationship between star formation and molecular gas change from as we move from entire galaxies to the individual environments. The empirical connection between molecular gas and star formation is well studied through the Kennicutt-Schmidt

relation (Kennicutt 1998), which links the surface density of gas Σ_{gas} , to the surface density of star formation Σ_{SFR} . ON galaxy integrated scales this relations holds a successful description on how these two quantities associate (Kennicutt 1998), (Schmidt 1959). However, because this relation is over integrated galaxy disks individual molecular gas clouds and stars at varying evolutionary stages are averaged over. Consequently many underlying physical properties of the galaxy are lost, concealing small scale physical processes. Recent surveys such as Physics at High-Angular Resolution in Nearby Galaxies (PHANGS) enables new scientific studies like these by providing a large sample of galaxies at 100 pc (Leroy et al. 2021). We utilized high-resolution measurements from the Multi Unit Spectroscopic Explorer (MUSE) and the Atacama Large Millimeter/sub-millimeter Array (ALMA) to link local SFR surface densities to gas mass surface densities (measured on ~ 100 parsec scales). In this work, we investigate the

Kennicutt-Schmidt relation in NGC 1566 across the full disk, and how this relation varies across different environments within the galaxy such as the bar, spiral arms, and interarm. In addition to comparing the differences in depletion time of the molecular gas. Depletion time is a complementary way to describe the efficiency of star formation. It is defined as : Equation 1:

$$\tau_{\text{dep}} = \frac{\Sigma_{\text{gas}}}{\dot{\Sigma}_{\text{SFR}}}, \quad (1)$$

and the timescale required for an amount of available gas to be converted into a star given the current star formation rate stays constant. Equivalently, the inverse depletion time can be used to measure the star formation efficiency such that $\text{SFE} \sim 1/\tau_{\text{dep}}$ (Bigiel et al. 2008), (Bigiel et al. 2011), (Leroy et al. 2013). Our second goal was to compare the slopes and scatter of each environment to empirically determine the various physical properties that are affecting each region independently. We investigated how the best-fit slope, normalization, and scatter of SFR for each region depended on estimations of physical processes in observable quantities like SFR and gas mass. Section 2 will introduce our data processes and how (Querejeta et al. 2021) created environmental masks for certain regions of NGC 1566, Section 3 will illustrate the techniques we implemented, Section 4 will analyze our slopes in relation to the original (Kennicutt 1998) findings and discuss each environment individually, Section 5 discusses the importance of our results and how this will impact future directions when studying star formation on large scales.

2. DATA

Our MUSE and ALMA data are both reprojected and convolved to the same pixel dimensions so that the pixel-to-pixel comparisons can be carried out. The astrometry and pixel dimensions are matched between the two datasets allowing us to view them on the same fragment of the sky. We also matched each dataset’s resolution via a Gaussian kernel, blurring the ALMA image to match MUSE at (0.8 arcseconds). In order to correct for the inclination of NGC 1566 which is at an angle of $i = 29.5^\circ$ we deprojected the datasets recovering the 2D shape to determine its true size and to understand its physical properties.

2.1. ALMA

The molecular properties of NGC 1566 were acquired from observations made by the ALMA as a part of the PHANGS-ALMA survey (Leroy et al. 2021). These observations trace molecular gas through emissions from carbon monoxide (CO), which serves as a proxy to mea-

sure the molecular hydrogen H_2 . H_2 is the main constituent in gas that is used in star formation however it is typically hard to observe under typical conditions in nearby galaxies. CO emission is commonly used to estimate the molecular gas surface density Σ_{mol} .

2.2. MUSE

Measurements of the star formation rate were observed using the MUSE as a part of the PHANGS-MUSE survey (Emsellem et al. 2022). MUSE is an optical integral-field spectrograph that provides spatially resolved spectroscopy across nearby galaxies. These observations allow the properties of ionized gas and young stellar populations to be measured across the disk of NGC 1566.

2.3. Environment Masks

Using environmental masks from Querejeta et al. (2021), we decomposed the galaxy into various environments and observed significant variation in the slope of the power-law relation. The PHANGS team sampled 74 nearby galaxies and constructed morphological maps of sub-galactic environments based on *Spitzer* 3.6 μm images. They distinguished 5 different environments, including the spiral arms, centres, bars, interarm regions, and discs without strong spirals; however, in our study, we mainly used masks for the spiral arms, bar, and interarm regions as this is where we observe the most SF to be the most concentrated in NGC 1566. We observed a slope of 1.04 ± 0.06 in the interarms, 0.36 ± 0.05 in the bar, and 0.78 ± 0.03 in the spiral arms.

3. METHODS

3.1. CO(2-1) conversion to Molecular Gas

For our work we used the CO(2-1) intensity integrated maps to derive the molecular gas surface density across the full disk. The high-resolution of this PHANGS-ALMA study allows us to observe physical properties on ~ 100 pc scales. The CO intensity is converted from $\text{Jy beam}^{-1} \text{km s}^{-1}$ to brightness temperature units K km s^{-1} using the standard Rayleigh-Jeans relation, accounting for the synthesized beam size and rest frequency of the observation. We convert the CO(2-1) intensity to a CO(1-0)-equivalent intensity assuming a line ratio of $R_{21} = 0.65$ (Leroy et al. 2021), and adopt a Milky Way-like CO-to- H_2 conversion factor of $\alpha_{\text{CO}} = 4.35 M_\odot \text{K km s}^{-1}$ (Bolatto et al. 2013) (including the helium correction). The gas surface density is then deprojected to a face-on value assuming an inclination of $i = 32^\circ$:

$$\Sigma_{\text{gas}} = \alpha_{\text{CO}} \times \frac{I_{\text{CO}}}{R_{21}} / \cos(i) \quad (2)$$

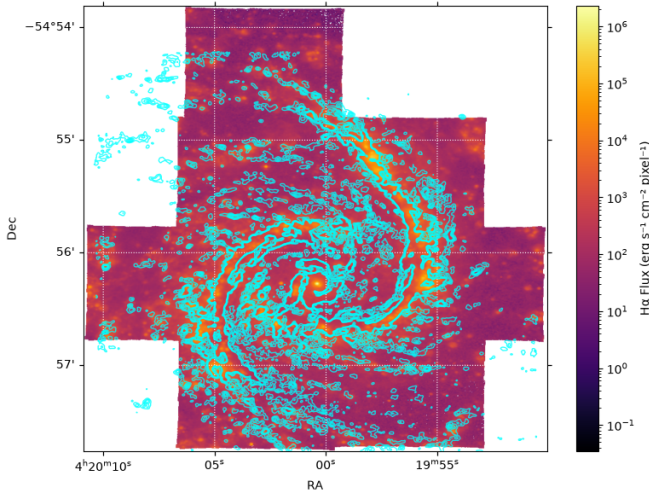


Figure 1. The Convolved MUSE map overlaid with CO(2-1) contours. The right-side y-axis shows units of $H\alpha$ flux, while the left-side y-axis is the dec of the galaxy’s position. The x-axis shows the RA. The importance of these maps is to help us visualize the CO(2-1) contour trace the $H\alpha$ emission, which is the primary prediction of the KS relation.

3.2. $H\alpha$ conversion to SFR

The $H\alpha$ measurements were corrected for dust attenuation/extinction before being converted into Σ_{SFR} . The resulting star formation rate surface density maps were analyzed at spatial scales comparable to the molecular gas observations, allowing us to investigate the relationship between Σ_{mol} and Σ_{SFR} on ~ 100 pc scales. Extinction correction is applied using the Balmer decrement with the color excess $E(B-V)$ converted to $H\alpha$ attenuation via $A(H\alpha) = k_{H\alpha} \times E(B-V)$, where $k_{H\alpha} = 2.53$ (Cardelli et al. 1989), (Hummer & Storey 1987). The dust-corrected $H\alpha$ flux in each pixel is converted to a luminosity assuming a distance of $D = 17.69$ Mpc (Pessa et al. 2021), and the corresponding star formation rate is calculated using the Kennicutt (1998) calibration, $SFR = 7.910^{-42} L(H\alpha)$.

3.3. SNR thresholds

To ensure statistically significant measurements we chose to create signal-to-noise SNR thresholds on both of our observed quantities. Only regions with emissions greater than our adopted thresholds are included in the KS analysis. Applying these thresholds allows us to reduce the influence of noise and uncertainty in our derived relation. Each threshold is derived independently due to each measurement having its own uncertainties and sensitivity (a product of the observation tools). We defined the gas detection threshold to be 3 times the

median CO rms noise:

$$\Sigma_{\text{gas},3\sigma} = \frac{3 \times \text{median}(\sigma_{\text{CO}})}{R_{21}} \times \frac{\alpha_{\text{CO}}}{\cos(i)} \quad (3)$$

. Similarly, the SFR detection threshold is defined as five times the median 1σ uncertainty in Σ_{SFR} , obtained by propagating the $H\alpha$ flux error map through the identical Σ_{SFR} conversion chain described above. Only pixels satisfying $\Sigma_{\text{gas}} > \Sigma_{\text{gas},3\sigma}$ and $\Sigma_{\text{SFR}} > \Sigma_{\text{SFR},5\sigma}$, in addition to an independent joint $H\alpha + \text{CO}$ $S/N \geq 3$ mask, are included in the KS relation fit.

3.4. Bayesian Hierarchical Linmix MCMC Fitting

The Kennicutt-Schmidt relation was fit in log-space using Bayesian Hierarchical regression linmix method (Kelly 2007). We model the relation as $\log(\Sigma_{\text{SFR}}) = N \log(\Sigma_{\text{mol}}) + b$ where N represents the slope and b represents the normalization. Both of our observed quantities contain uncertainties, linmix accounts for errors in both independent and dependent variables. The fitting procedure also includes a intrinsic scatter variable σ_{int} that accounts for the deviation of the data lying around the KS relation beyond the reported measurement uncertainties. This Bayesian framework uses MCMC sampling to explore the parameter space defined by N , b , σ_{int} . At each iteration, the likelihood of a proposed set of parameters is evaluated based on how well the corresponding linear relation describes the observed $\log(\Sigma_{\text{mol}})$ and $\log(\Sigma_{\text{SFR}})$ measurements while accounting for their associated uncertainties. The resulting MCMC samples form posterior distributions for the fitted parameters, from which we determine the best-fit slope, intercept, intrinsic scatter, and their associated uncertainties. We ran the linmix sampler for 5000 iterations and used the converged posterior distribution to characterize the KS relation.

4. RESULTS

4.1. Global Results

Following our high-resolution pixel-to-pixel comparison of $H\alpha$ and CO intensities in NGC 1566 2, we found $N = 0.92 \pm 0.02$. This relation is consistent with findings in other high-resolution studies like Leroy et al. (2013), Bigiel et al. (2008) and Pessa et al. (2021). Bigiel et al. (2008), Leroy et al. (2013) and Pessa et al. (2021) have found strong linear relationships between molecular gas and star formation even when individual galaxies were spatially resolved however, the increase in scatter and variation observed between local environments points to the fact that the KS relation does not imply a universal star formation rate. Instead local variations in molecular gas clouds and stellar evolution may influence

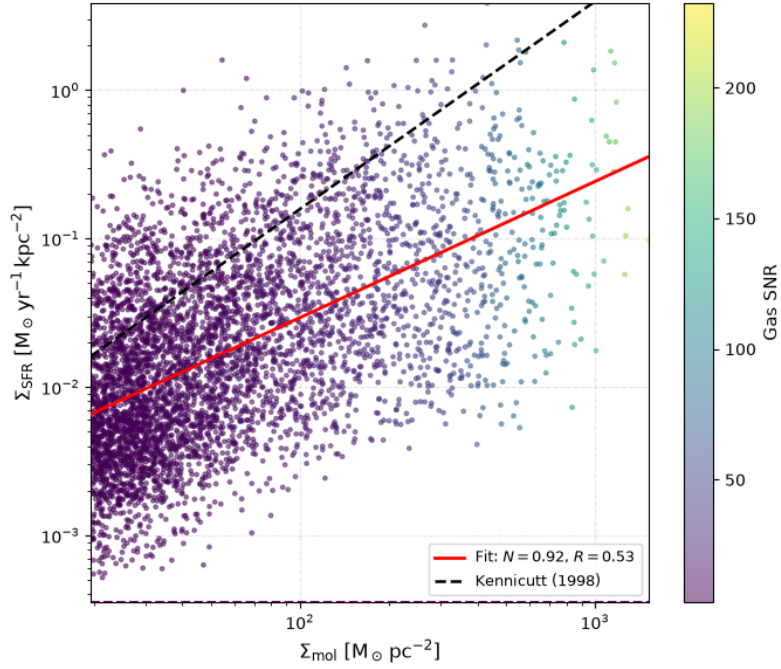


Figure 2. KS relation for NGC 1566 across the galaxy’s entire disk. The solid red line indicates the power-law relation at $N = 0.92 \pm 0.02$, compared with the accepted Kennicutt (1998) slope of 1.4. This shows a zoomed-in view where x and y are confined to their respective threshold values, with a color bar indicating the SNR of the molecular gas. We observe that when we conduct high-resolution observations of NGC 1566 as opposed to integrated values across many different galaxies, we obtain a power-law relation that is systematically lower than the accepted Kennicutt (1998) slope

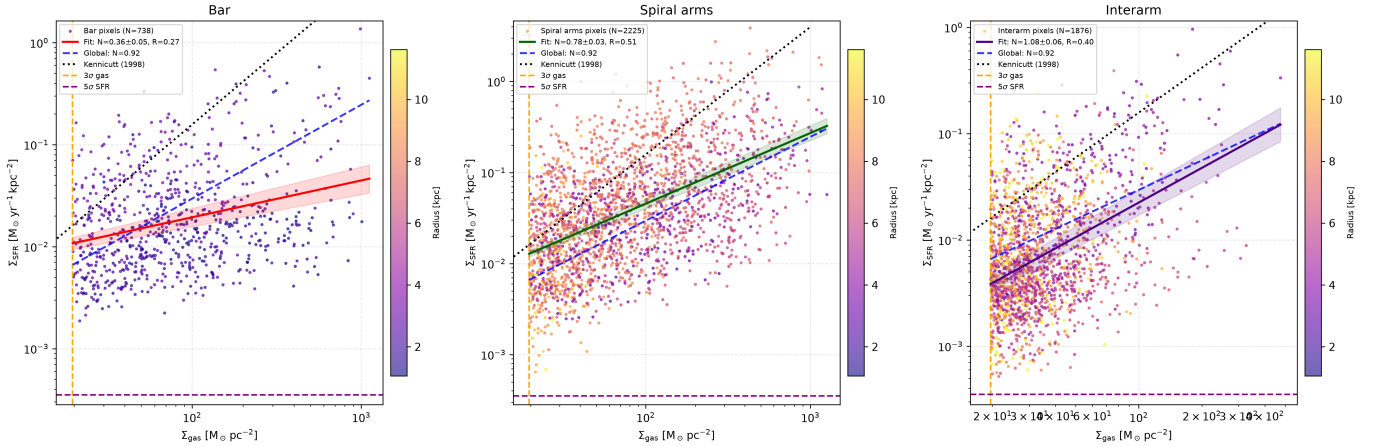


Figure 3. The panel above displays each environment in comparison to the global slope of the galaxy indicated by the blue dashed line. Like the previous figure, the black dashed line shows the accepted Kennicutt slope. We observe that the relation between Σ_{SFR} and Σ_{gas} varies significantly based upon environmental conditions; the bar shows the highest deviation from our global slope, while we see a tighter correlation between the interarms and the global slope.

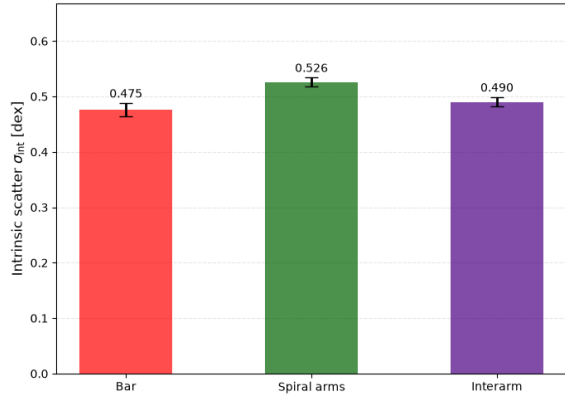


Figure 4. Intrinsic scatter σ_{int} of the Kennicutt-Schmidt relation measured independently for each morphological environment in NGC 1566 using the LinMix Bayesian linear regression. The bar (red), spiral arms (green), and interarm (purple) regions show similar intrinsic scatter values of 0.475, 0.526, and 0.490 dex, respectively, despite showing significantly different KS slopes and normalizations. Error bars represent the 16th–84th percentile uncertainty on σ_{int} derived from the posterior MCMC chain. The uniformity of scatter across environments suggests that pixel-to-pixel variations in star formation efficiency are driven by local physical processes operating independently of large-scale morphological structure.

the normalization, slope, and scatter of the KS relation. We find that in NGC 1566 this is exactly the case with local environments showing varying slopes, normalization, and scatter. Furthermore, our global fit sits systematically below the [Kennicutt \(1998\)](#) relation by approximately 0.5 dex in normalization, consistent with the finding that galaxy-integrated measurements weight toward the brightest star-forming regions and overestimate the global star formation efficiency when compared to resolved pixel-to-pixel analyses.

4.2. Environment: Bar

The bar showed the most deviation from the global slope, as NGC 1566 is a Type 1 Seyfert Active Galactic Nucleus (AGN) host galaxy. We expected this result, as it is known that bar quenching has a cyclical relationship with AGN-hosting galaxies ([Liu & Zhou 2026](#)). Strong gravitational forces created by the bar’s rotation cause large quantities of cold gas from the disk to funnel in towards the supermassive black hole (creating the AGN) and fuel centralized star formation. [Pessa et al. \(2021\)](#) described the bar region of the sampled galaxies to be the “star formation desert” as their studies show a smaller range of SFR in the bars, all systematically below the bar sequence. Despite these large differences in normalization and slope across environments, the intrinsic scatter in the KS relation remains consis-

tent, measuring 0.475 dex in the bar, 0.526 dex in the spiral arms, and 0.490 dex in the interarm. This uniformity in scatter suggests that pixel-to-pixel variations in star formation efficiency are governed by local physical processes operating independently of large-scale morphological structure, even as the mean efficiency varies enormously between environments.

4.3. Environment: Spiral Arms

We observed the spiral arms to have a steeper slope than the bar, indicating that as more gas is introduced into the environment, star formation is more efficient. This result is expected as spiral arms typically have a more conducive environment for forming stars. Spiral arms act as a sort of “cosmic highway” for interstellar materials flowing in and out of the galaxy. Density waves are compact regions of the “cosmic highway” where there are large build-ups of stellar material, providing perfect conditions for intense star formation, therefore leading to a steeper slope in the KS relation ([Sun et al. 2023](#)). This finding is consistent with [Pessa et al. \(2021\)](#) and [Querejeta et al. \(2021\)](#) as they both report a higher average $\Sigma_{mol,gas}$ and Σ_{SFR} in the spiral arms however, our study does not show a higher average in the spiral arms in comparison to the interarm region. We see that at lower radii, the spiral arms have a higher average of $\Sigma_{mol,gas}$ and Σ_{SFR} compared to the interarm region (consistent with [Pessa et al. \(2021\)](#) and [Querejeta et al. \(2021\)](#)), and reverse at a larger radius. The bar quenching may be a factor in the star formation efficiency of the spiral arms as well, because the buildup of gas near the bar ends acts as a funnel exit into the “cosmic highway,” providing denser regions of gas. This shows that in the spiral arms we can more accurately predict the gas-to-star formation rate, likely due to the density waves compressing gas and triggering star formation.

4.4. Environment: Interarms

We observed the interarm region to have the steepest slope of the three environments. This was a surprising result, as we expected star formation efficiency to increase around dense regions of gas while the interarm typically has more diffuse gas. This result could be due to several factors, one being a result of our sigma threshold. We chose to cut off low signal points; for Σ_{gas} we cut off points below 5σ and for Σ_{SFR} points below 3σ . However, because the interarm region is diffuse and gas is largely spread out, it lacks uniformity. This could cause us to sample a certain region of the interarm that happens to have a denser region of gas in comparison to portions of the environment that have low signal.

Another intrinsic factor could be that we are sampling more regions that have converted HI to H_2 . As Σ_{gas} increases, more gas becomes available for star formation, so our slope is partially showing that transition of HI to H_2 across our dynamical range rather than pure star formation.

5. CONCLUSION

We studied the power-law relation between Σ_{SFR} and Σ_{gas} for the galaxy NGC 1566. By following studies such as Leroy et al. (2021), Bigiel et al. (2008), and Pessa et al. (2021) we conducted a high-resolution study that looks into the complexities of the Kennicutt-Schmidt relation first discovered and broadly researched in 1998. The initial study by Kennicutt (1998) determined the global average rate of star formation and discovered a tight correlation resulting in a power-law slope of 1.4. However, as seen in Leroy et al. (2021), Bigiel et al. (2008), and Pessa et al. (2021) a universal law is not sufficient. When you resolve the sub-kpc (cloud-scale) interstellar medium, we observe different results and how molecular gas and star formation interact. The Kennicutt (1998) study created a broad, sweeping view by comparing two ends of the spectrum with normal disk

galaxies (low - moderate density) and starbursts (high density). Our study analyzed one specific galaxy (NGC 1566), showing that at high-resolution, the KS relation varies where N is approximately 1 and differs based on the local environment within NGC 1566. These variations across the bar, spiral arms, and interarm sections suggest that the KS relation is not consistent across the full disk and is dependent on the local dynamics and unique physical conditions of said environment. From these results, we observed that the Kennicutt-Schmidt relation changes when looking at individual environments within a galaxy; the relation between Σ_{SFR} and Σ_{gas} is not uniform across the full disk when observed at high-resolutions. The distinction between these two studies is important as their measurements are significantly different; however, it highlights the innate differences within a single galaxy's environments. Our environmental decomposition results indicate to us that, the KS relation in NGC 1566 is neither universal nor spatially uniform — the slope, normalization, and scatter all vary significantly between morphological environments and with galactocentric radius, underscoring that the global KS relation measured by (Kennicutt 1998) represents an average over physically distinct regimes that behave in fundamentally different ways at the 100 pc scales.

REFERENCES

- Bigiel, F., Leroy, A., Walter, F., et al. 2008, *The Astronomical Journal*, 136, 2846
- . 2011, *The Astrophysical Journal Letters*, 730, L13
- Bolatto, A. D., Wolfire, M., & Leroy, A. K. 2013, *Annual Review of Astronomy and Astrophysics*, 51, 207
- Cardelli, J. A., Clayton, G. C., & Mathis, J. S. 1989, *Astrophysical Journal*, Part 1 (ISSN 0004-637X), vol. 345, Oct. 1, 1989, p. 245-256., 345, 245
- Emsellem, E., Schinnerer, E., Santoro, F., et al. 2022, *Astronomy & Astrophysics*, 659, A191
- Hummer, D., & Storey, P. 1987, *Monthly Notices of the Royal Astronomical Society* (ISSN 0035-8711), vol. 224, Feb. 15, 1987, p. 801-820. SERC-supported research., 224, 801
- Jeřábková, T., Hasani Zonoozi, A., Kroupa, P., et al. 2018, *Astronomy & Astrophysics*, 620, A39
- Kelly, B. C. 2007, *The Astrophysical Journal*, 665, 1489
- Kennicutt, Jr, R. C. 1998, *The astrophysical journal*, 498, 541
- Leroy, A. K., Walter, F., Sandstrom, K., et al. 2013, *The Astronomical Journal*, 146, 19
- Leroy, A. K., Hughes, A., Liu, D., et al. 2021, *The Astrophysical Journal Supplement Series*, 255, 19
- Liu, J., & Zhou, Z. 2026, *The Astrophysical Journal*, 1003, 25
- Ostriker, E. C., & Shetty, R. 2011, *The Astrophysical Journal*, 731, 41
- Pessa, I., Schinnerer, E., Belfiore, F., et al. 2021, *Astronomy & Astrophysics*, 650, A134
- Querejeta, M., Schinnerer, E., Meidt, S., et al. 2021, *Astronomy & Astrophysics*, 656, A133
- Schmidt, M. 1959, *Astrophysical Journal*, vol. 129, p. 243, 129, 243
- Sun, J., Leroy, A. K., Ostriker, E. C., et al. 2023, *The Astrophysical journal letters*, 945, L19